Illustration 10.4

Examine the differentiability of $f(x) = x^{\frac{1}{3}}$ at $x = 0$.

Solution

Let $f(x) = x^{\frac{1}{3}}$. Clearly, there is no hole (or break) in the graph of this function and hence it is continuous at all points of its domain.

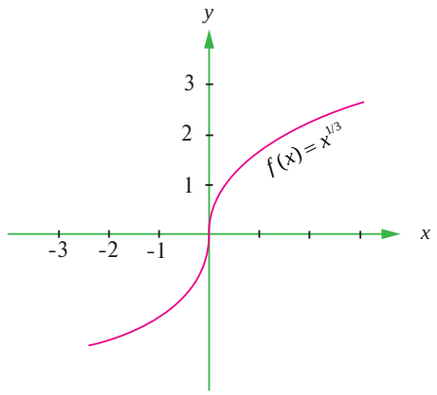


Fig. 10.19

Let us check whether $f'(0)$ exists.

Now $$f'(0) = \lim_{x\to 0}\frac{f(x)-f(0)}{x-0} = \lim_{x\to 0}\frac{x^{\frac{1}{3}}-0}{x}$$

$$= \lim_{x\to 0} x^{\frac{-2}{3}} = \lim_{x\to 0}\frac{1}{x^{\frac{2}{3}}} \to \infty$$

Therefore, the function is not differentiable at $x = 0$. From the Fig. 10.19, further we conclude that the tangent line is vertical at $x = 0$. So f is not differentiable at $x = 0$.

If a function is continuous at a point, then it is not necessary that the function is differentiable at that point.

Example 10.3

Show that the greatest integer function $f(x) = \lfloor x \rfloor$ is not differentiable at any integer?

Solution

The greatest integer function $f(x) = \lfloor x \rfloor$ is not continuous at every integer point n, since $\lim_{x\to n^-} \lfloor x \rfloor = n-1$ and $\lim_{x\to n^+} \lfloor x \rfloor = n$. Thus $f'(n)$ does not exist.

What can you say about the differentiability of this function at other points?

Illustration 10.5

Let $f(x) = \begin{cases} x & x \le 0 \\ 1+x & x > 0 \end{cases}$

Compute $f'(0)$ if it exists.

Solution

Look at the graph drawn.

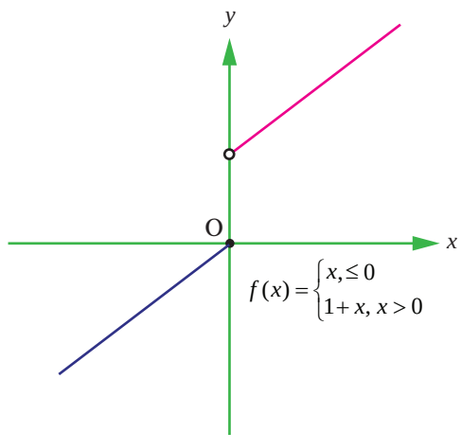


Fig. 10.20

$$f'(0^-) = \lim_{\Delta x\to 0^-}\frac{f(0+\Delta x)-f(0)}{\Delta x}$$

$$= \lim_{\Delta x\to 0^-}\frac{f(\Delta x)}{\Delta x}$$

$$f'(0^-) = \lim_{\Delta x\to 0^-}\frac{\Delta x}{\Delta x} = 1$$

$$f'(0^+) = \lim_{\Delta x\to 0^+}\frac{1+\Delta x}{\Delta x}$$